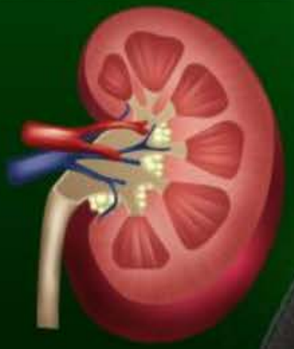
VICTORY 2027

The Nervous System



LECTURE 2 | BIOLOGY

Upasana Ma'am



Topics To Be Covered

1. Transmission of the Nerve Impulse
2. Types of Nerves, Types of Nervous System

TYPES OF NEURONS

SIGNALS PASS THROUGH AND BETWEEN NEURONS

↓

Information is collected by the dendritic tips

↓

The chemical reaction sets off electric impulse

↓

Electric impulses pass from dendrite to cell body

↓

Then passes from cell body through the axon

↓

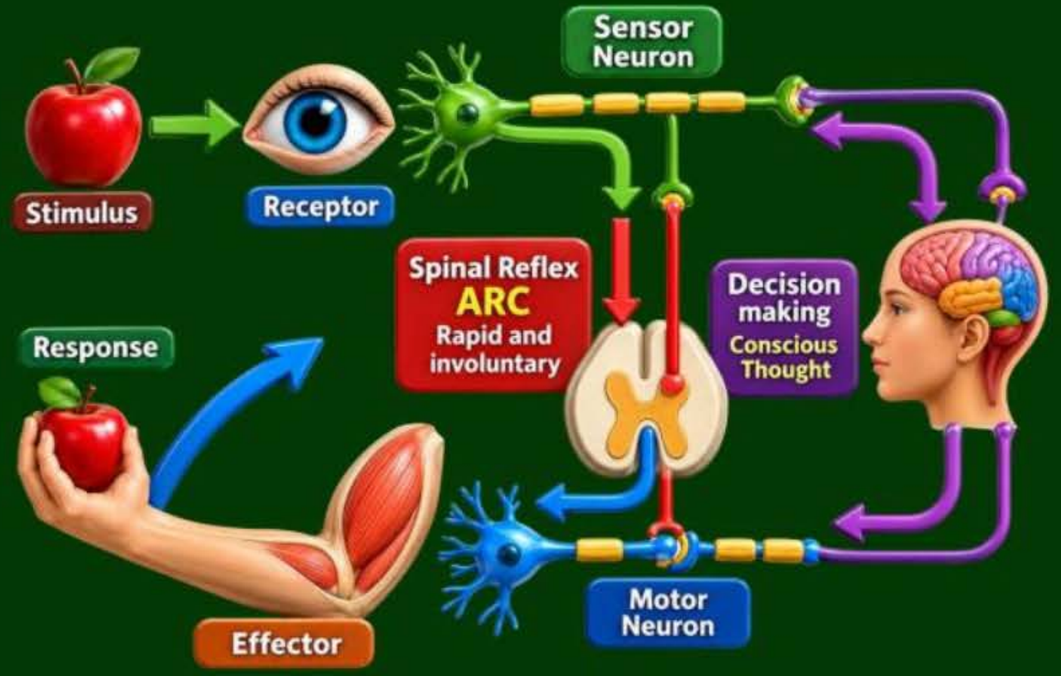
Reaches nerve endings at the end of axon

↓

Electric impulse sets off signals called neurotransmitters across the synapse

↓

This again passes as electric signals in the next dendrite and so on

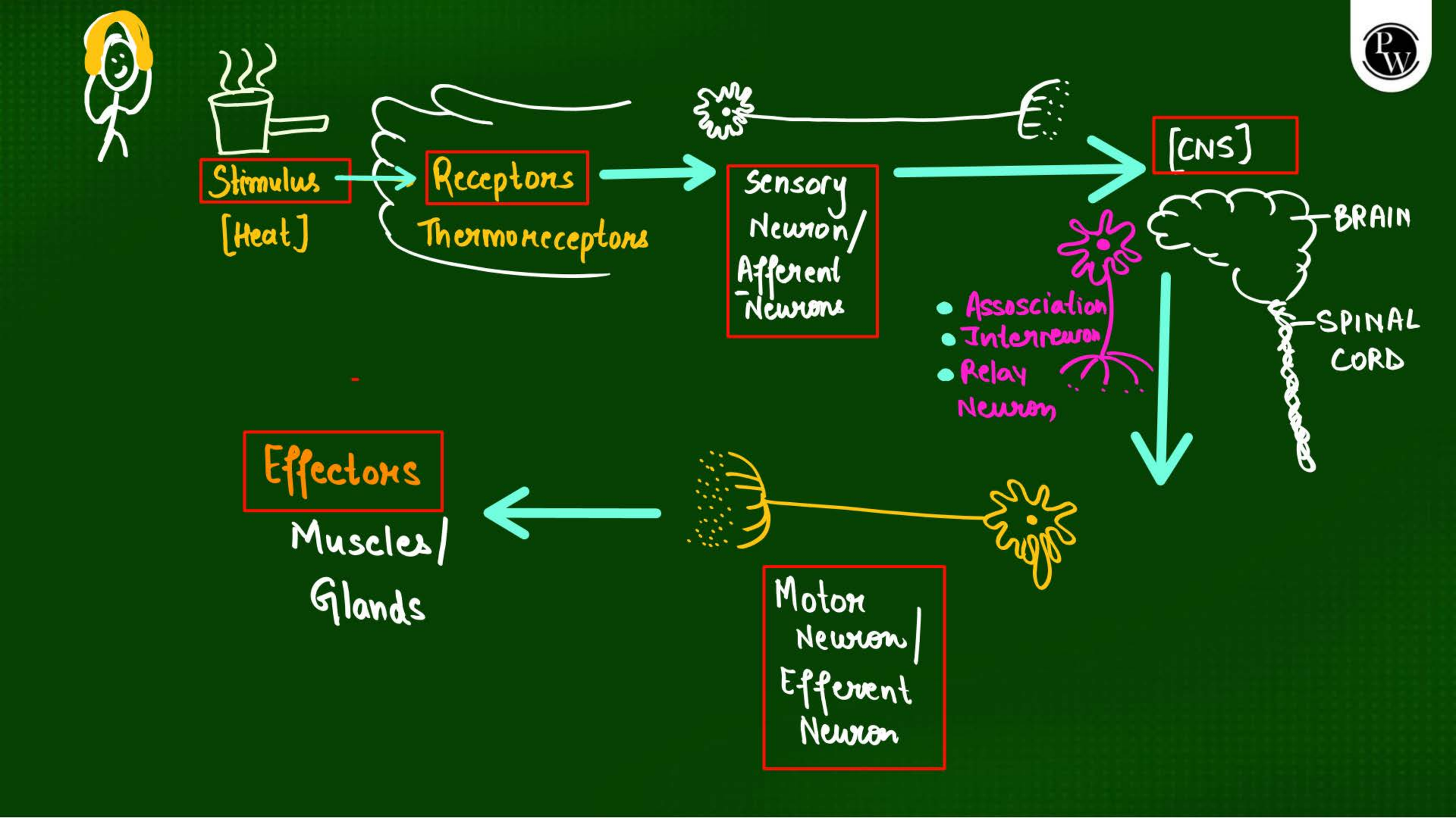


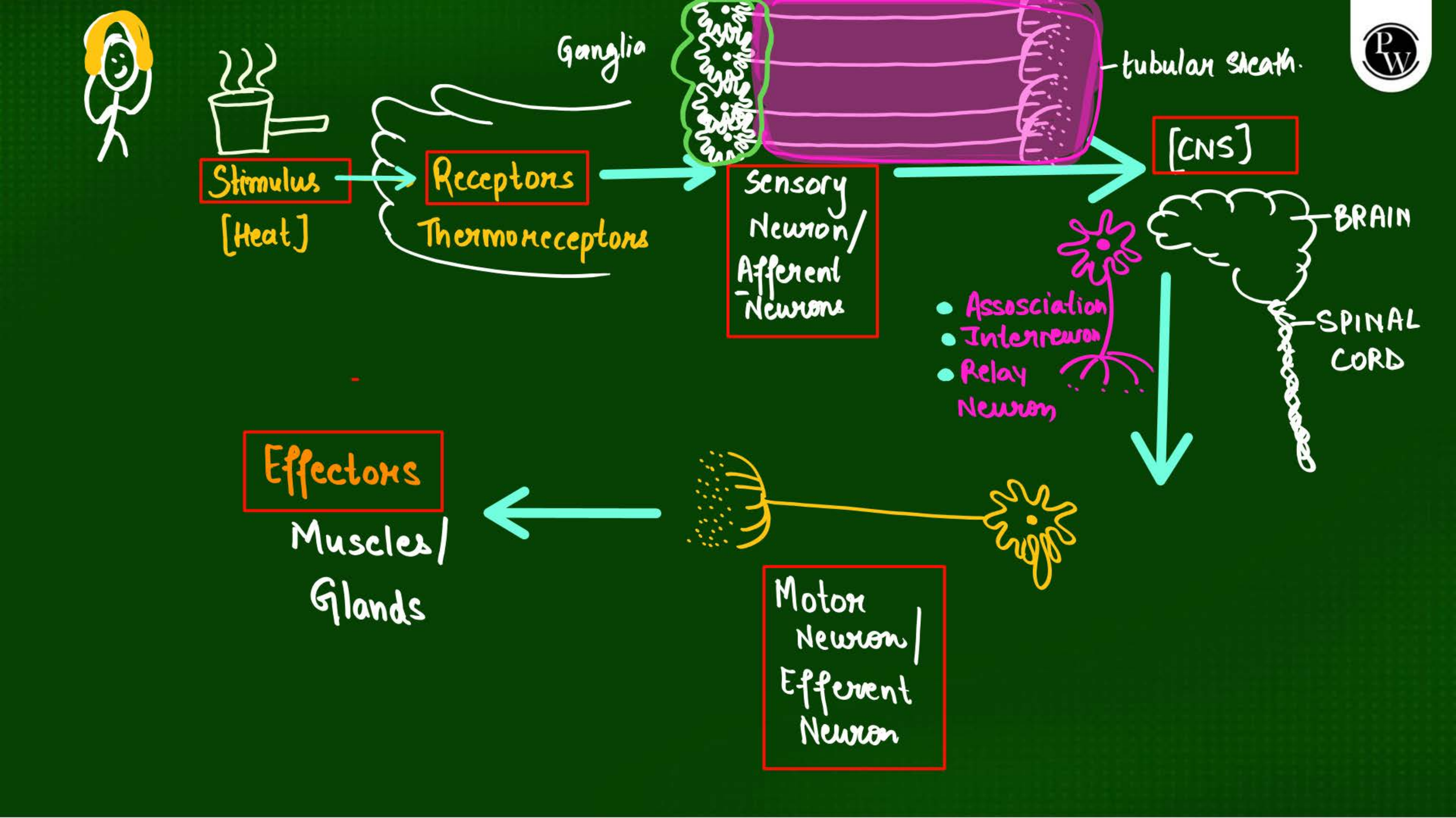
Stimuli → Any external or internal agent of the environment that brings the response.

'Receptors'

Special cells which are responsible for receiving the stimuli

- Photoreceptors → light
- Phonoreceptors → sound
- Thigmoreceptors → Touch
- Thermoreceptors → Temperature
- Olfactory receptors → Smell
- Gustatory receptors → Taste


Stimulus
[Heat]
Receptors
Thermoreceptors
Sensory Neuron/ Afferent Neurons
[CNS]
BRAIN
SPINAL CORD
Association Interneuron Relay Neuron
Effectors
Muscles/ Glands
Motor Neuron/ Efferent Neuron


Ganglia
tubular sheath
Stimulus
[Heat]
Receptors
Thermoreceptors
Sensory Neuron/ Afferent Neurons
[CNS]
BRAIN
SPINAL CORD
Assosciation Interneuron
Relay Neuron
Effectors
Muscles/ Glands
Motor Neuron/ Efferent Neuron

QUESTIONS

The basic structural and functional unit of brain is

A Cyton

B Nephron

C Axon

D Neuron

QUESTIONS

Which of the following structures is absent in nerve cells?

A Nucleus

B Cytoplasm

C Centrosome

D Neurilemma

QUESTIONS

Nissl's granules are found in the cytoplasm of:

A Nerve cell

B Dendrons

C Axon

D Dendrites

QUESTIONS

Any factor that can bring a change in the body of an organism is called :

A Response

B Stimulus

C Reflex

D Impulse

From Relay Neuron to Effector ____ is required.

A) Brain
B) Sensory Neuron
C) Motor Neuron ✓
D) Stimulus

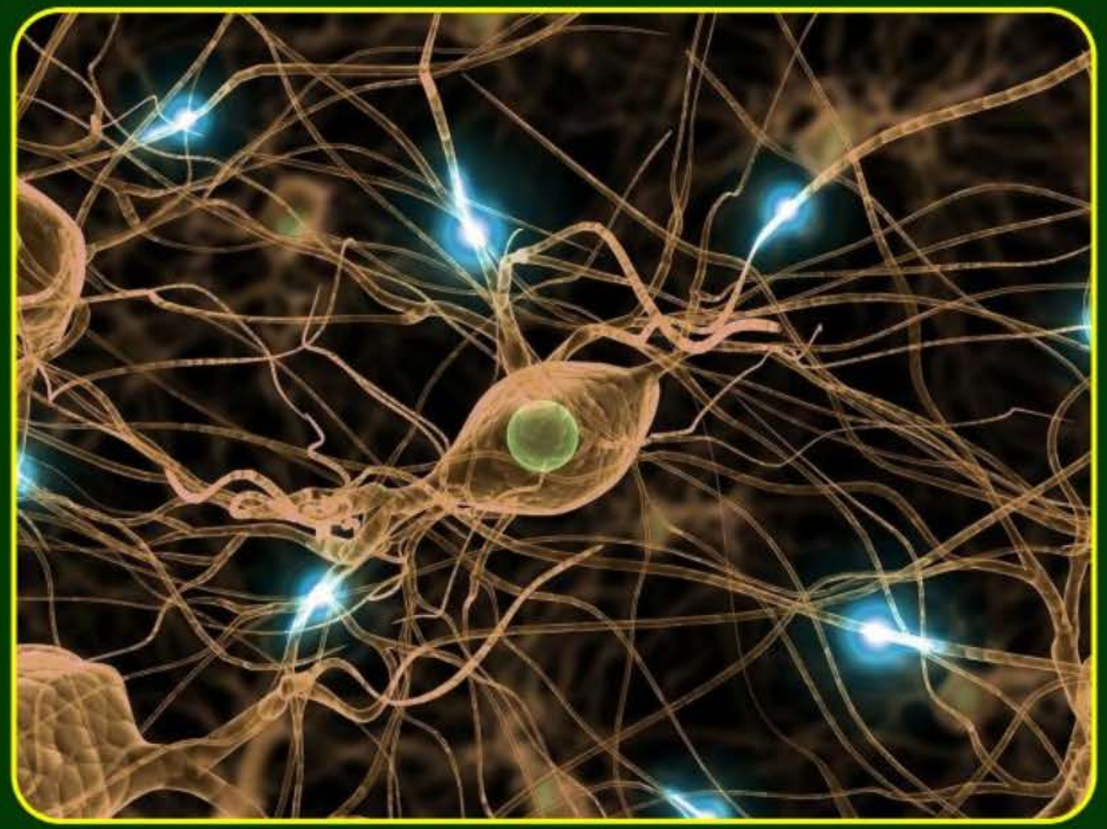
NERVES

axon fibre.

- Nerve is a bundle of nerve fibres (axons) of separate neurons, enclosed in a tubular sheath.

myelin

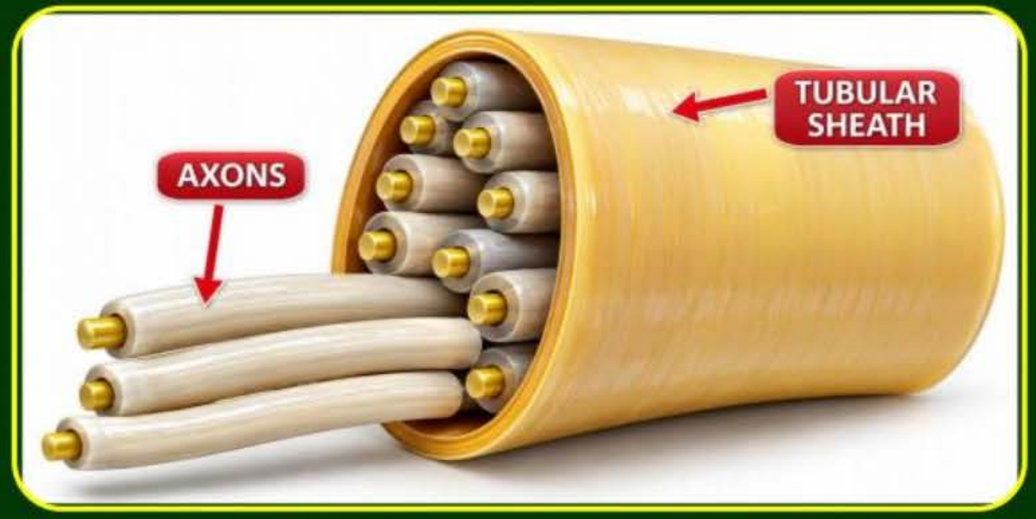
- Nerves are the thread-like white structures which emerge from the brain and spinal cord and branch out to almost all parts of the body.



THREE KINDS OF NERVES ARE AS FOLLOWS

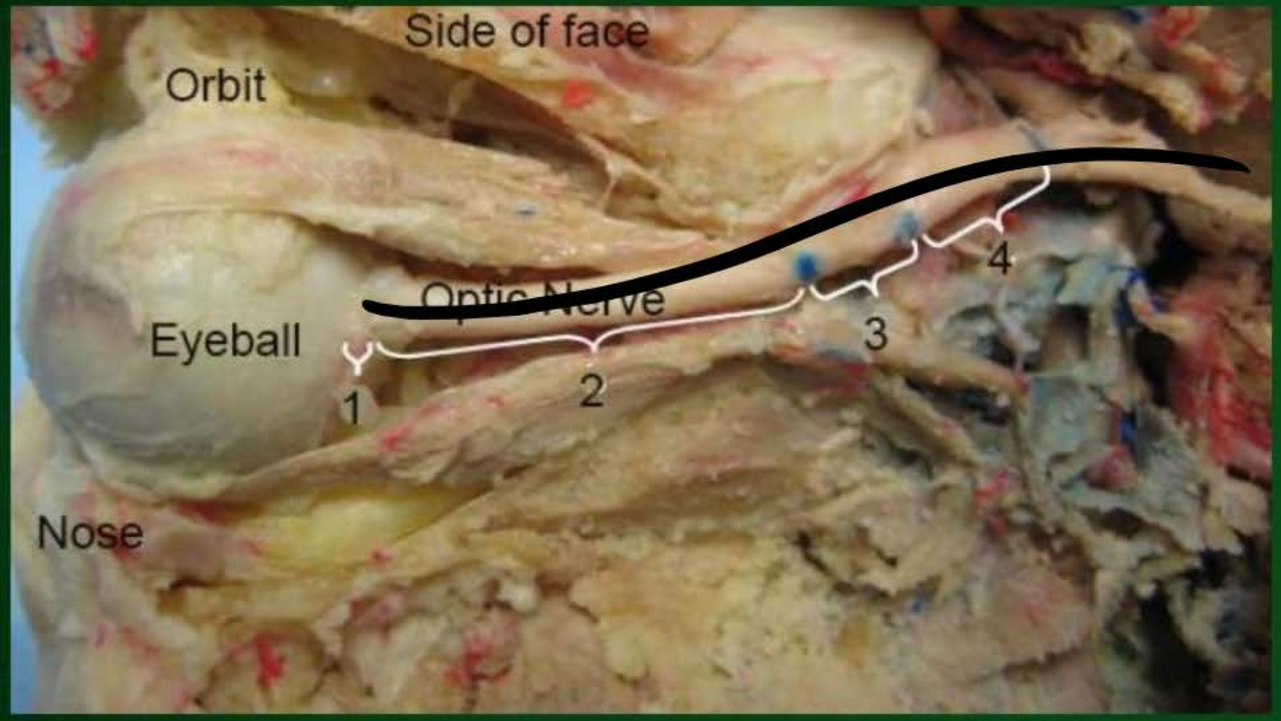
1. Sensory nerves : Sensory nerves contain only sensory fibres bringing impulses from the receptors (sense organs) to the brain or spinal cord.

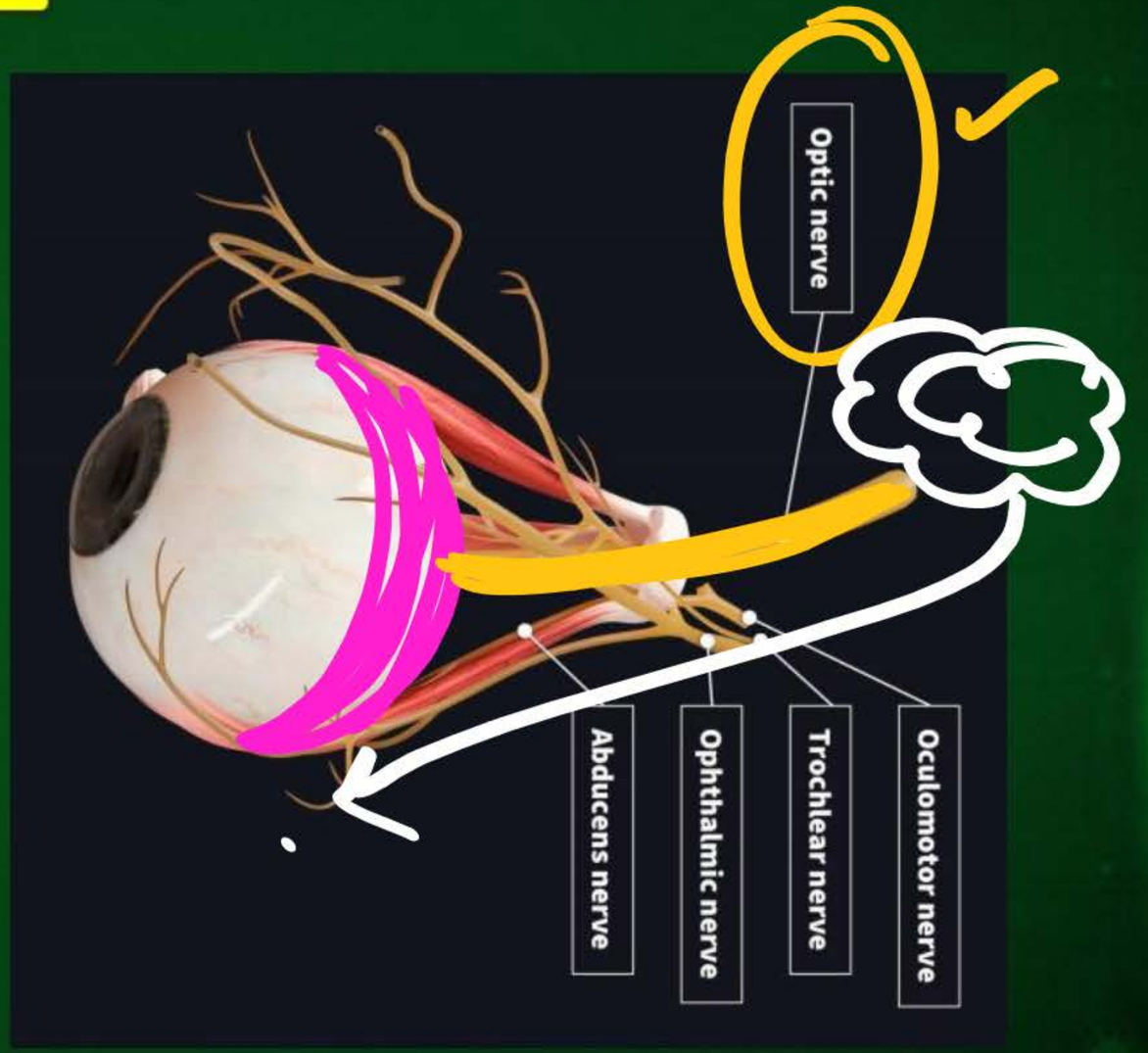
Example: Optic nerve arising from the eye and ending in the brain.



A Nerve, formed of a bundle of axons

THREE KINDS OF NERVES ARE AS FOLLOWS

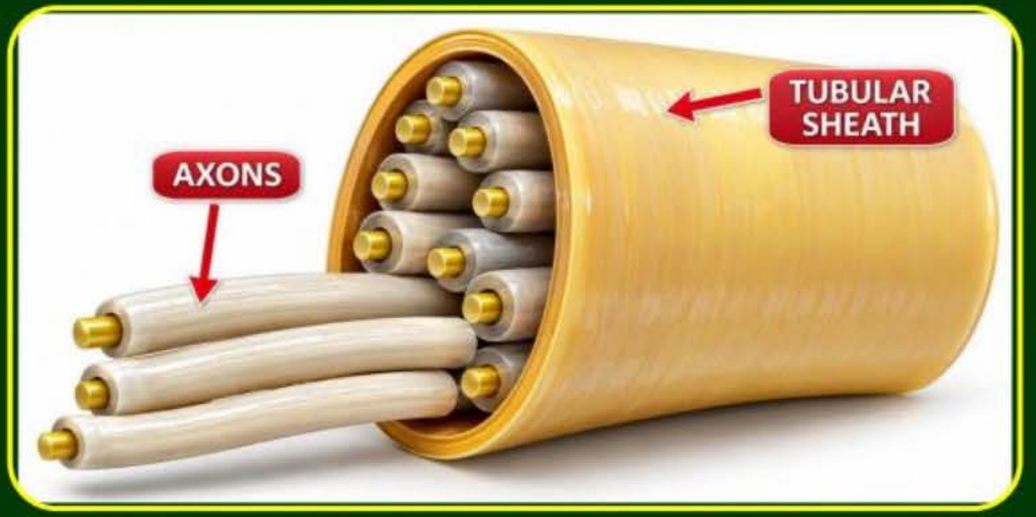




THREE KINDS OF NERVES ARE AS FOLLOWS

2. Motor nerves : Contain only motor fibres carrying impulses from the brain or spinal cord to effector organs (muscles or glands) to bring them into action.

Example: A nerve arising from the brain and supplying the muscles of the eyeball for rotating the eye.



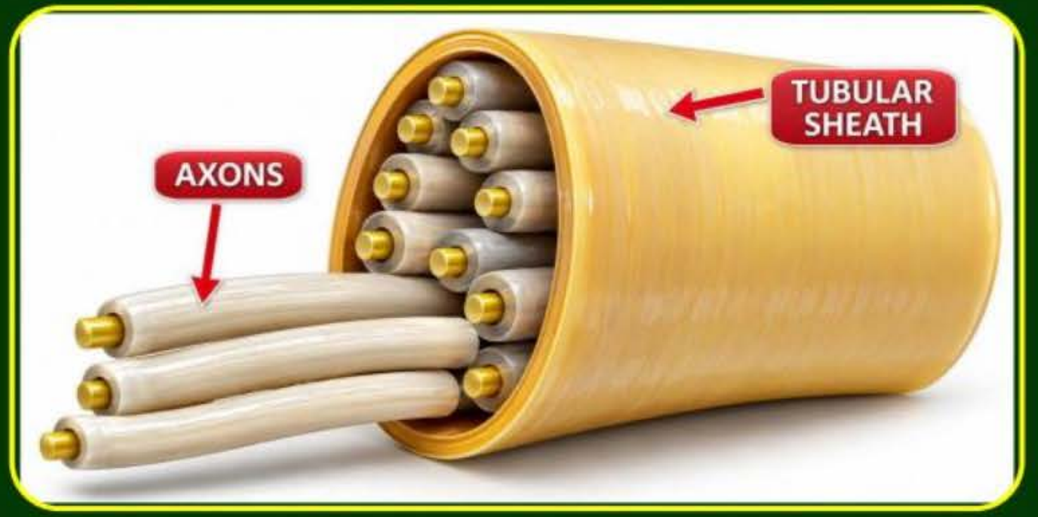
A Nerve, formed of a bundle of axons

THREE KINDS OF NERVES ARE AS FOLLOWS

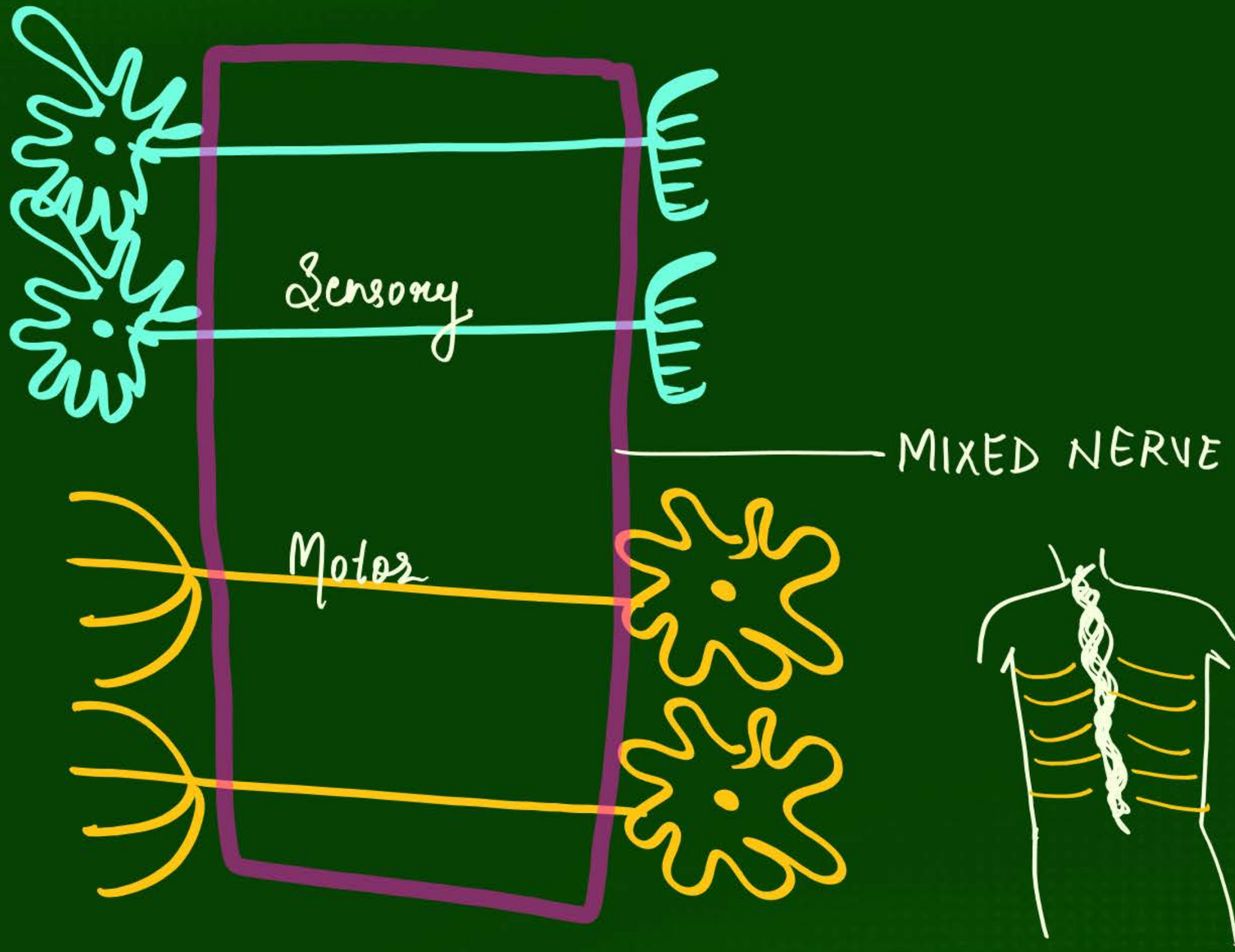
3. **Mixed nerves** are those, that contain both sensory and motor fibres.

Example: A spinal nerve.

- **Ganglia (sing. ganglion)** are the aggregates of the cytons or cell bodies of nerve cells from which the nerve fibres may arise or enter into.



A Nerve, formed of a bundle of axons


Sensory
Motor
MIXED NERVE

A Nerve which gives command to Effector after receiving from CNS —

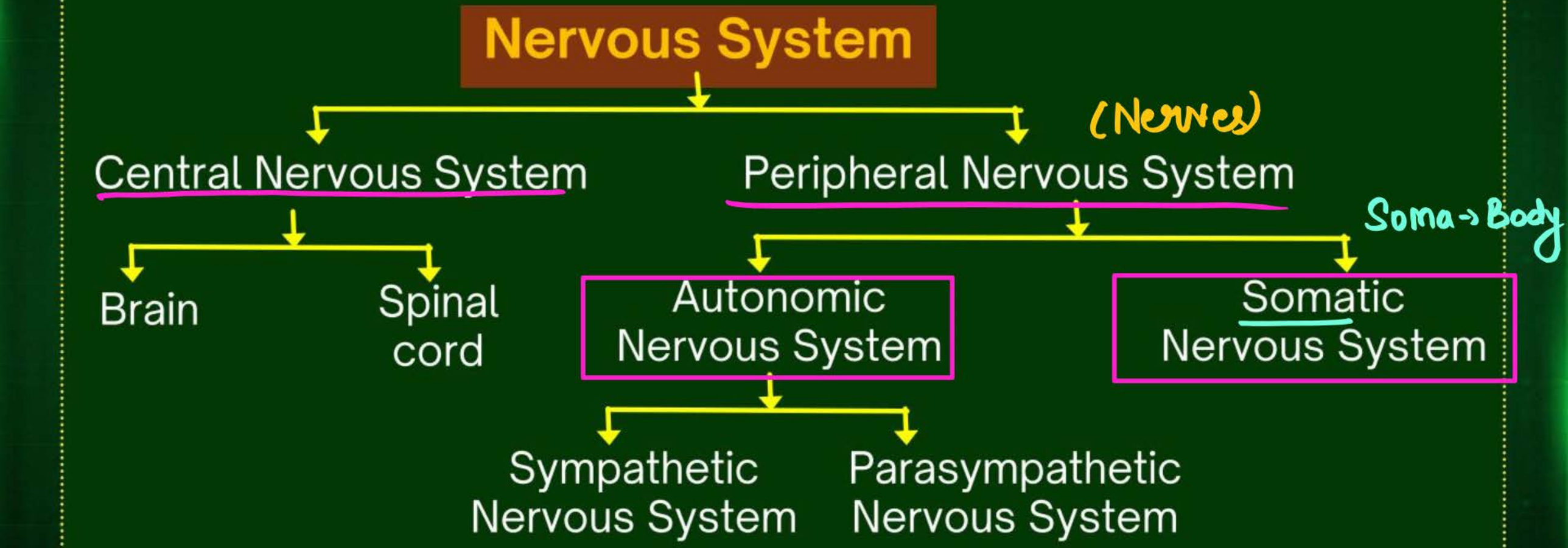
A) MIXED

B) SENSORY

C) MOTOR ✓

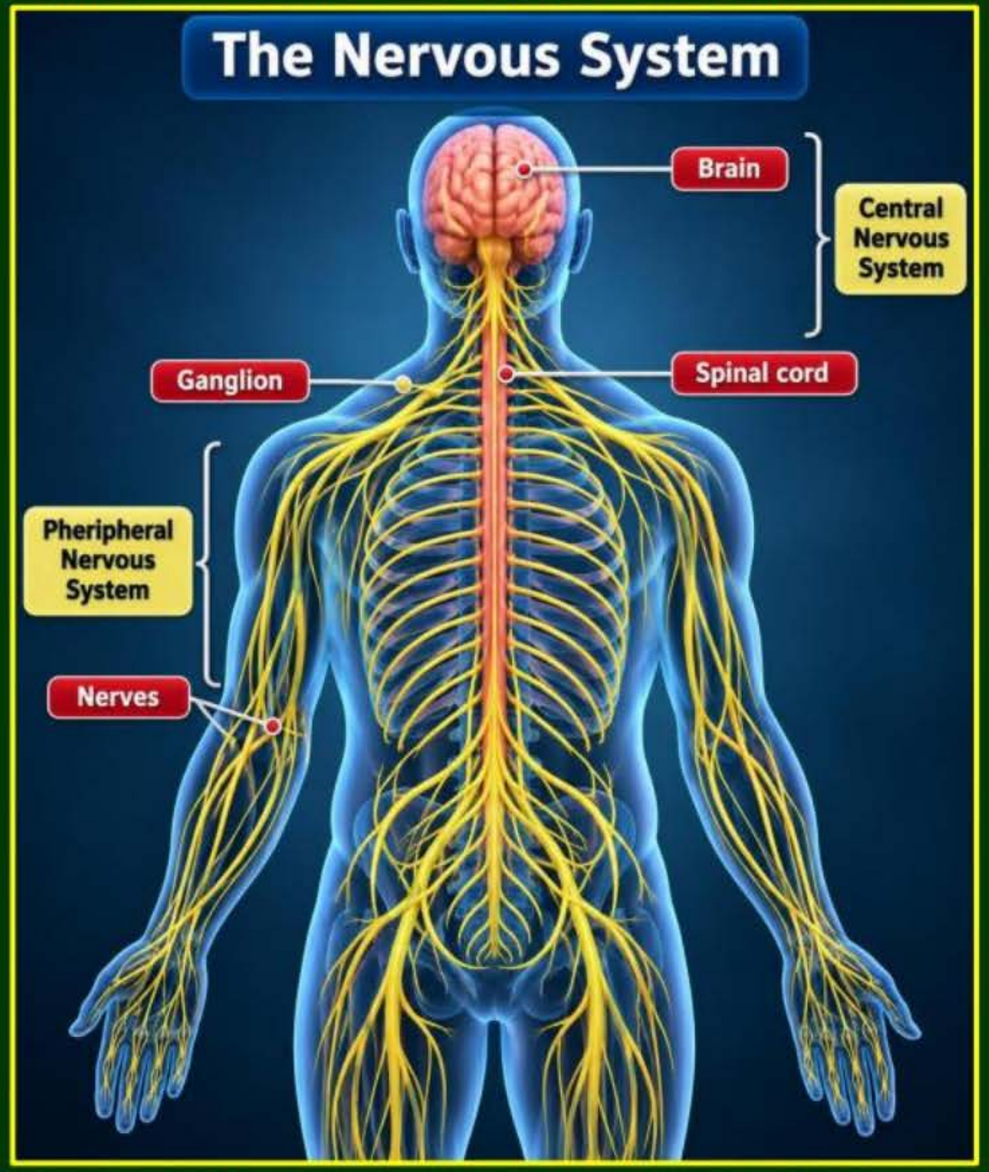
D) SUNDAR HOGI

THREE KINDS OF NERVES ARE AS FOLLOWS



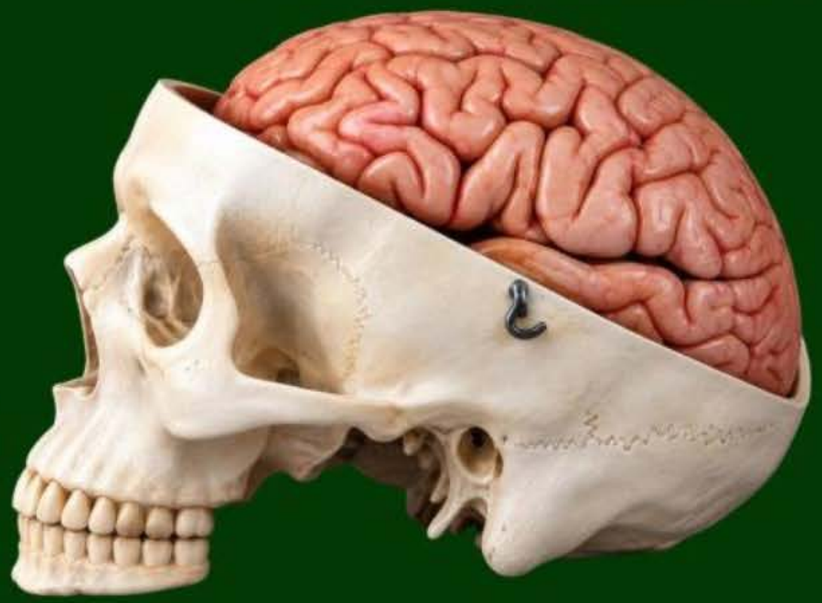
CENTRAL NERVOUS SYSTEM (CNS)

1. Central nervous system (CNS) includes the brain and the spinal cord contained within the vertebral column.

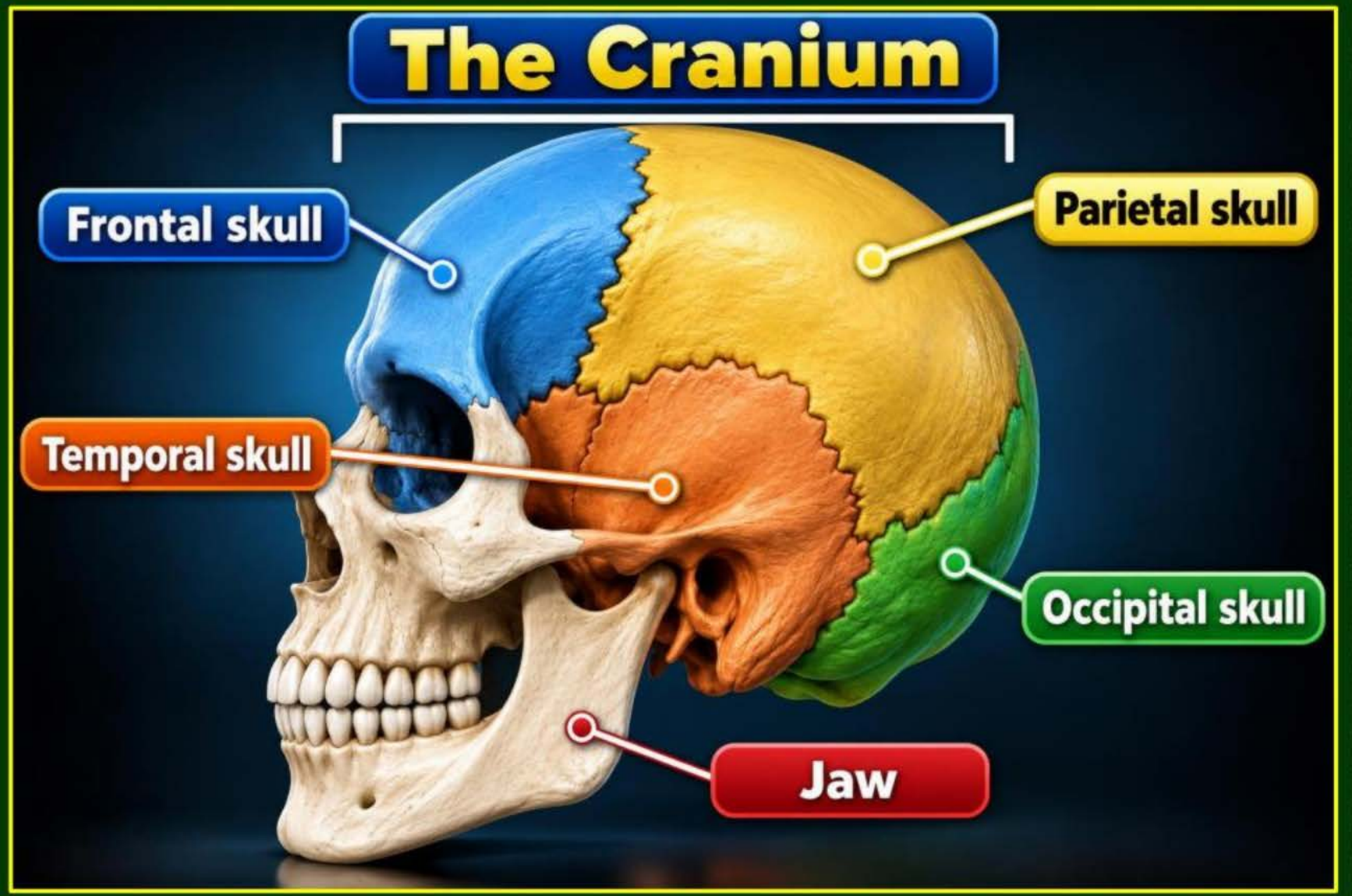


THE BRAIN

- Largest among all animals.
- Weighs about 1.35 kg and constitutes about 2% of the total body weight.
- Protected by CRANIUM or Bony skull.



THE BRAIN


The Cranium
Frontal skull
Parietal skull
Temporal skull
Occipital skull
Jaw

THE BRAIN

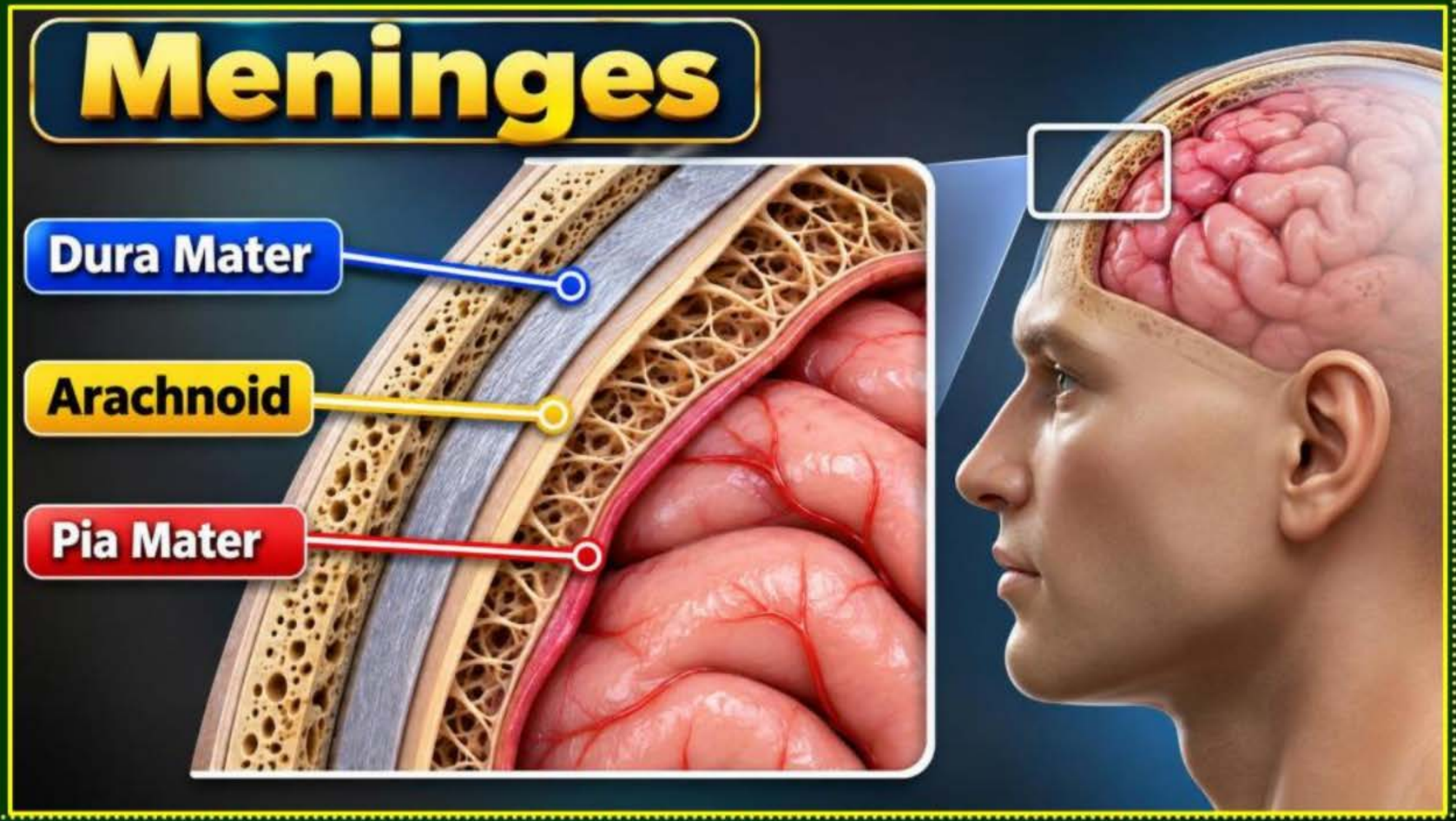


- **Coverings (Meninges) :** The brain is protected by 3 membranous coverings called meninges (meninx : membrane) which continue backwards on the spinal cord.

- It is 80% water and consumes more than 25% (one fourth) of the total oxygen taken into the body.

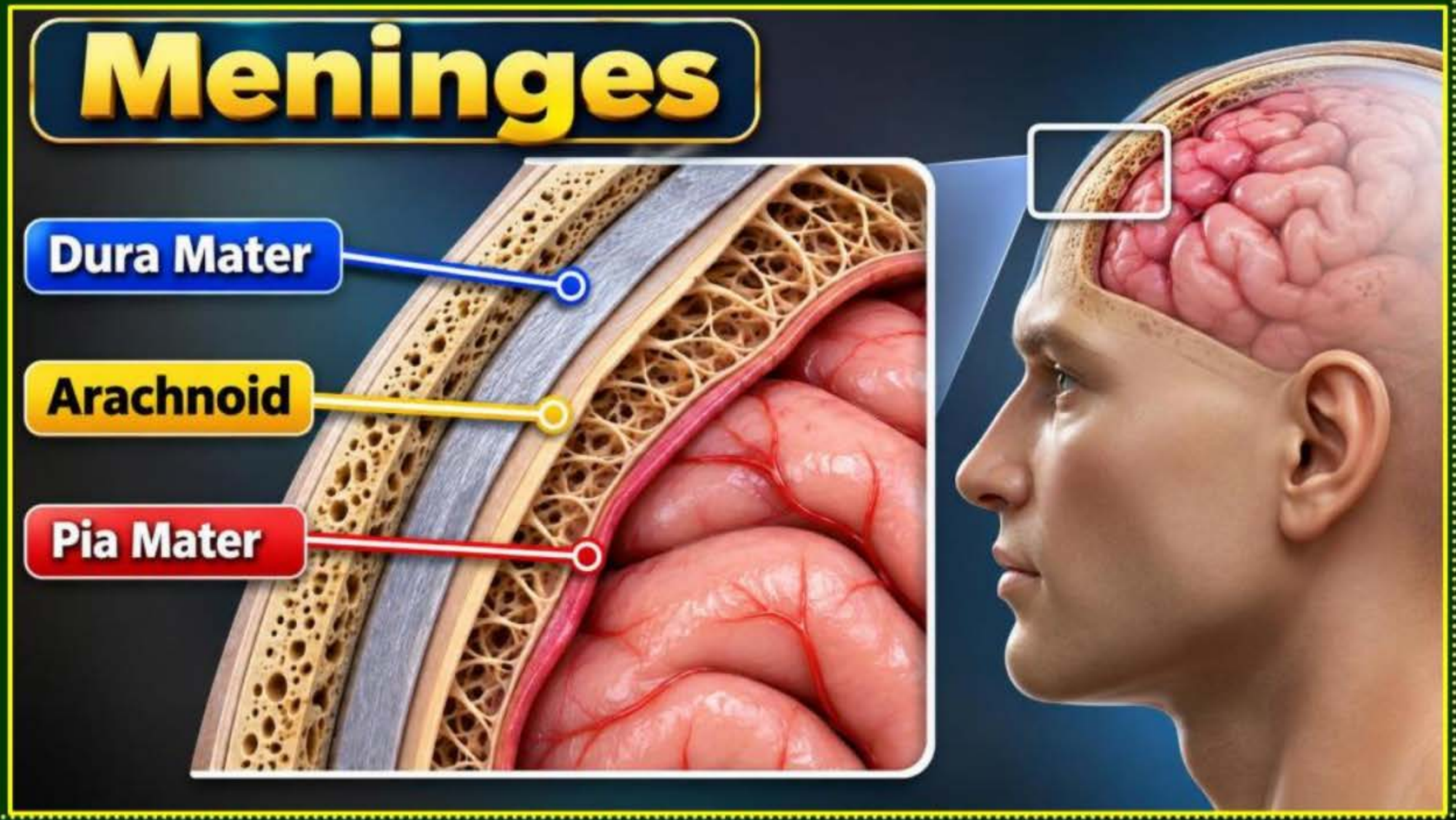
THE BRAIN

(i) Dura mater - the outermost tough fibrous membrane (dura : tough, mater: mother).



THE BRAIN

(ii) Arachnoid - the thin delicate middle layer giving a web-like cushion (arachne: spider).



THE BRAIN

(iii) Pia mater - the innermost highly vascular membrane, richly supplied with blood (pia : tender).

Meningitis is the inflammation of the meninges.

The space between the covering membranes is filled with a watery fluid –

Cerebrospinal fluid (CSF) which acts like a cushion to protect the brain from shocks.

THE BRAIN



THE BRAIN

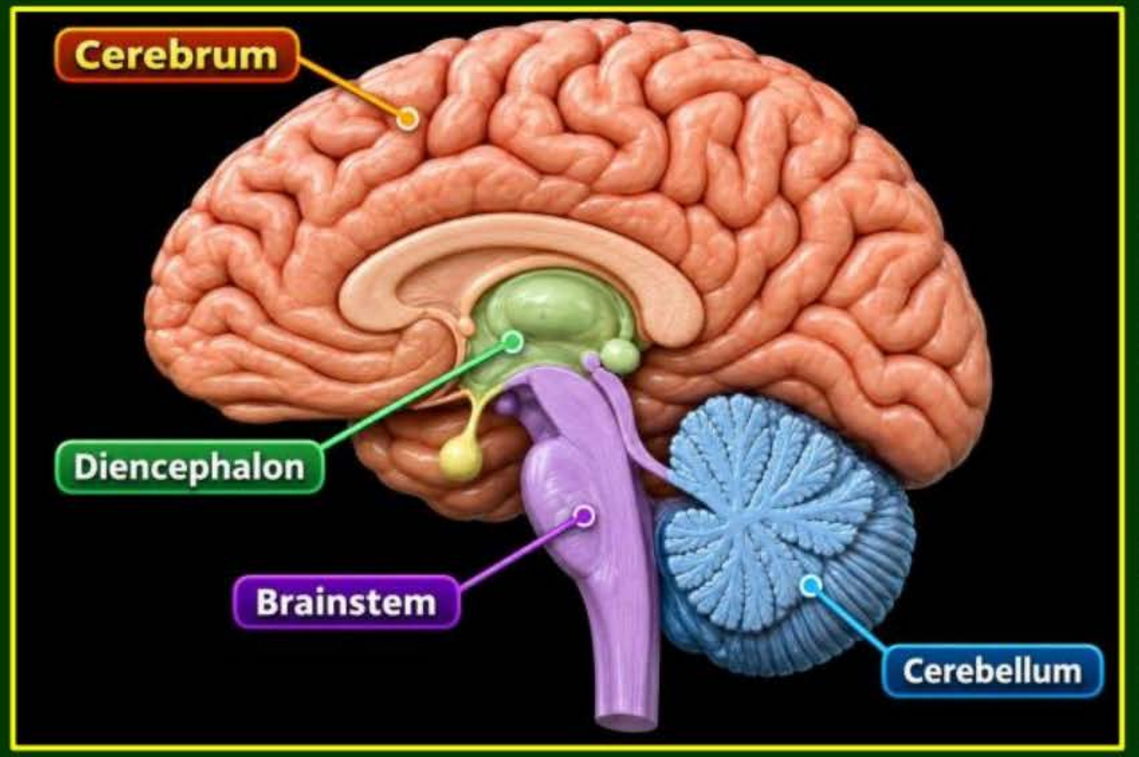
Three Primary Regions of the Brain

- A. Forebrain
  - 1. Cerebrum
  - 2. Diencephalon
    - Thalamus
    - Hypothalamus
- B. Midbrain
- C. Hindbrain
  - 1. Cerebellum
  - 2. Pons
  - 3. Medulla oblongata

FOREBRAIN

Diencephalon

- (a) Thalamus (relays pain and pressure impulses to cerebrum).
- (b) Hypothalamus (controls the body temperature and pituitary).



Thank You